3.1.3 Transport in plants

As plants become larger and more complex, transport systems become essential to supply nutrients to, and remove waste from, individual cells.

The supply of nutrients from the soil relies upon the flow of water through a vascular system, as does the movement of the products of photosynthesis.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for transport systems in multicellular plants	To include an appreciation of size, metabolic rate and surface area to volume ratio (SA:V). *M0.1, M0.3, M0.4, M1.1, M2.1, M4.1* HSW1, HSW3, HSW5, HSW8
(b)	**(i)** the structure and function of the vascular system in the roots, stems and leaves of herbaceous dicotyledonous plants	To include xylem vessels, sieve tube elements and companion cells.
	(ii) the examination and drawing of stained sections of plant tissue to show the distribution of xylem and phloem	**PAG1** HSW4
	(iii) the dissection of stems, both longitudinally and transversely, and their examination to demonstrate the position and structure of xylem vessels	**PAG2** HSW4
(c)	**(i)** the process of transpiration and the environmental factors that affect transpiration rate	To include an appreciation that transpiration is a consequence of gaseous exchange.
	(ii) practical investigations to estimate transpiration rates	To include the use of a potometer. *M0.1, M0.2, M1.1, M1.2, M1.3, M1.6, M1.11, M3.1, M3.2, M3.3, M3.5, M3.6, M4.1* **PAG5, PAG11** HSW2, HSW3, HSW4, HSW5, HSW6, HSW8
(d)	the transport of water into the plant, through the plant and to the air surrounding the leaves	To include details of the pathways taken by water **AND** the mechanisms of movement, in terms of water potential, adhesion, cohesion and the transpiration stream. HSW2, HSW8
(e)	adaptations of plants to the availability of water in their environment	To include xerophytes (cacti and marram grass) and hydrophytes (water lilies). HSW2

(f)	the mechanism of translocation.	To include translocation as an energy-requiring process transporting assimilates, especially sucrose, in the phloem between sources (e.g. leaves) and sinks (e.g. roots, meristem) **AND** details of active loading at the source and removal at the sink. HSW2, HSW8